

- 1 (a) A truck is accelerated from rest by a constant driving force. It experiences a drag force which is proportional to its speed. Eventually, the truck travels at a constant velocity.

Explain why the truck reaches a constant velocity.

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..... [2]

Due to the acceleration by the constant driving force, **truck's speed increases,** B1  
**hence the drag force** which opposes the driving force **increases in magnitude.**

**Eventually the drag force become equal in magnitude to the constant** B1  
**driving force.** Hence net force becomes zero, acceleration equals zero, hence constant velocity.

- (b) On Fig. 1.1 sketch a graph to show how the acceleration of the truck varies with time from the time when it was at rest to the time when it reaches the constant velocity.

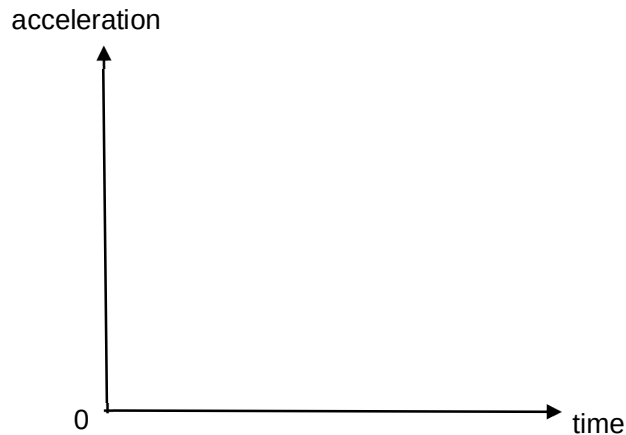
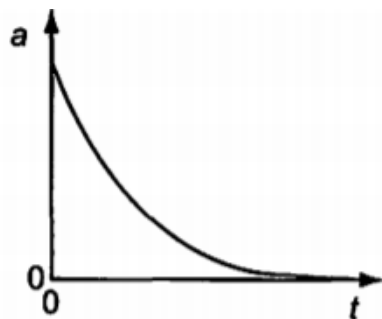


Fig. 1.1

[1]

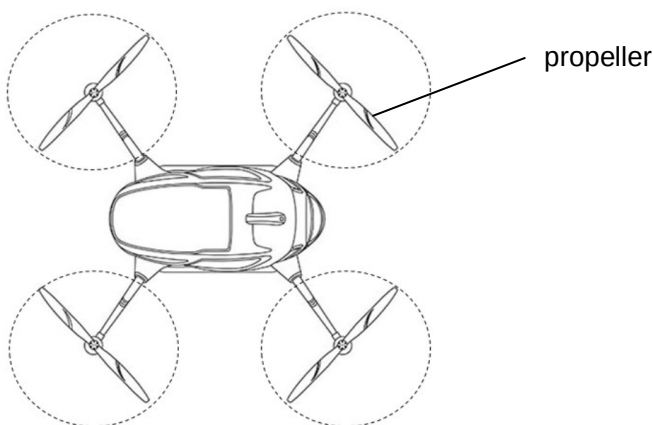
**Solution:**

B1



- acceleration decreases and reaches zero
- acceleration decreasing at a decreasing rate from start to the end

- (c) Fig. 1.2 shows a drone that is able to hover by rotating the four horizontal propellers attached to its body. The propellers push the air downwards and impart the air with a certain speed. The mass of the drone is 0.50 kg and each of the propellers has a radius of 15.0 cm.



Top view of a drone

**Fig. 1.2**

- (i) Calculate the upward force by the air on the drone.

upward force = ..... N [1]

**For equilibrium,**

**Upward force = Downward force =  $mg = 0.50 \times 9.81 = 4.905 = 4.9 \text{ N}$  A1**

- (ii) State, with a reason, the magnitude of the downward force exerted by all of the drone's propellers on the air.

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[2]

According Newton's 3<sup>rd</sup> Law, the downward force of the propellers on the air must be of the same magnitude as the upward force of the air on the propellers. M1

Total downward force = 4.9 N

A1

(iii) The density of air is  $1.3 \text{ kg m}^{-3}$ .

Determine using Newton's second law of motion the speed imparted to the air by the propellers.

speed = .....  $\text{m s}^{-1}$  [4]

L2

Assume propellers push air downward with a speed  $v$ .

Total cross sectional area of the air (for 4 propellers)

$$= 4 \times \pi r^2 = 4 \times \pi (0.15)^2 = 0.2827 \text{ m}^2$$

[Alternatively, divide the total downwards force of 4.905 N by 4]

Mass of air being pushed per unit time

$$= A \rho v$$

$$= 0.2827 \times 1.3 \times v$$

$$= 0.3676 v$$

M1

Momentum gained by air per unit time

$$= 0.3676 v \times \Delta v$$

$$= 0.3676 v \times (v - 0)$$

$$= 0.3676 v^2$$

M1

Rate of change of momentum of air =  $0.3676 v^2$

By Newton's 2<sup>nd</sup> Law,

Total downwards force on air by 4 propellers

= Rate of change of momentum of air

M1

$$4.905 = 0.3676 v^2$$

$$v = 3.654 = 3.7 \text{ m s}^{-1}$$

A1